

Artificial Intelligence for Virtual Reality: State of the Art, Challenges, and Future Perspectives

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AI has numerous applications in virtual reality (VR), particularly in analyzing the progress of its implementation, identifying existing limitations, and determining future development paths. This study finds that AI's main contributions to VR are in four key areas: creating intelligent virtual characters, advancing education and training, providing medical assistance, and generating dynamic scenes. This interdisciplinary convergence also brings significant opportunities, particularly in sectors such as education, healthcare, gaming, and corporate training. Additionally, the study discusses technical challenges related to computational costs, real-time feedback, user privacy, and algorithmic ethics. Critical challenges persist in data processing bottlenecks, privacy protection issues, and user adaptation. While AI enhances VR's intelligence and interactivity, breakthroughs are still needed in cross-modal integration, privacy, security, and user experience. Future advancements in deep learning and reinforcement learning may unlock unlimited potential for AI-driven VR in personalized adaptation and immersive interaction. Therefore, this study provides a comprehensive analysis of AI-VR integration, providing valuable insights for academic research and technological development.

CCS Concepts: • **Computing methodologies** → **Computer vision**; • **Human-centered computing** → **Virtual reality**;

Additional Key Words and Phrases: Artificial Intelligence, Virtual Reality, Intelligent Interaction, Digital Twins, User Experience, Privacy Security

ACM Reference format:

Yahui Wang, Mohsen Guizani, and M. Shamim Hossain. 2025. Artificial Intelligence for Virtual Reality: State of the Art, Challenges, and Future Perspectives. *ACM Trans. Multimedia Comput. Commun. Appl.* 21, 11, Article 327 (November 2025), 29 pages.

<https://doi.org/10.1145/3769090>

The authors extend their appreciation to the Deputyship for Research and Innovation, Ministry of Education in Saudi Arabia, for funding this research under grant number (IFKSU-HCRA-13-1).

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ACM 1551-6865/2025/11-ART327

<https://doi.org/10.1145/3769090>

1 Introduction

Virtual reality (VR) technology [1] has made breakthrough advancements in recent years, with its applications extending to multiple domains, including entertainment, education, healthcare, and military simulation. The construction of immersive digital spaces in VR revolutionizes **human-computer interaction (HCI)** and creates new development pathways across industries [2, 3]. However, as VR environments grow increasingly complex and interactive, conventional development approaches encounter significant bottlenecks in delivering truly intelligent, adaptive, and personalized user experiences. This has prompted researchers to integrate AI technologies [4] to overcome these limitations, making AI-driven VR a dynamic and rapidly expanding interdisciplinary field.

The integration of AI and VR encompasses a wide range of domains, from intelligent virtual characters to AI-assisted medical diagnostics within VR environments. Supported by **machine learning (ML)** frameworks and **deep learning (DL)** architectures, AI enables VR systems to analyze user behavior patterns [5]. **Computer vision (CV)** captures human motion trajectories, **natural language processing (NLP)** decodes speech information, and the fusion of multimodal data enables personalized environmental adaptation solutions [6]. Although a substantial body of research has explored specific applications of AI in VR, many of these studies remain isolated, focusing on niche applications (e.g., specific algorithms for single tasks) or on individual technical challenges [7]. Recent systematic reviews have similarly focused on specific domains, such as healthcare or education, or on integration with other technologies, including the **Internet of Things (IoT)** [8, 9]. As a result, there is a clear gap in the current literature—a lack of a comprehensive and multidimensional review that systematically connects cutting-edge technologies, diverse cross-domain applications, and global challenges, including technical bottlenecks, data privacy, algorithmic ethics, and user adaptability. Existing research generally lacks a macro-level synthesis to guide academic inquiry and technological advancement in this emerging field.

This study is thus motivated by the need for a holistic and integrative analysis of the rapidly evolving AI–VR domain. Its primary objective is to provide a comprehensive examination of AI–VR integration from a broader perspective, offering valuable insights for both academic research and technological development. To achieve this goal, the study aims to:

- Systematically review the key AI technologies in VR and their functional roles, constructing a coherent conceptual framework;
- Map the main application areas, ranging from education and healthcare to gaming and corporate training, and identify pioneering use cases;
- Analyze persistent technical, ethical, and user-centered challenges that hinder technology adoption and immersive experience enhancement;
- Offer forward-looking perspectives on potential developmental trajectories in AI–VR integration, promoting the evolution toward more mature and innovative solutions.

By identifying current limitations and emerging opportunities, this study seeks to outline the potential growth pathways for AI–VR integration and provide practical guidance for professionals, thereby bridging the gap between theory and practice.

The structure of this study is organized as follows: Section 1 introduces the development background of VR technology and the important role of AI within it while also explaining the research motivation and objectives. Section 2 provides a detailed overview of the technical foundations of AI in VR and its applications in different fields. Section 3 reviews the current research status of AI–VR integration, highlighting its progress in various fields, including education, gaming, and healthcare. Section 4 explores the challenges and future directions of combining AI with VR in detail. Finally, Section 5 summarizes the main findings and conclusions of the study.

2 Technical Framework and Core Applications of AI and VR Integration

VR technology is now a major player in the tech world. Improving user experience and redefining interaction models are key to making it a mainstream solution. AI, as a core driver of tech leaps, helps supercharge VR with its strengths in data analysis, pattern recognition, and smart decision-making. With AI, virtual worlds become smarter, and users enjoy more lifelike and personalized digital experiences. Figure 1 presents an intuitive overview of the AI–VR integration framework, centered on the core process of “perception–decision–generation.” The powerful synergy between AI and VR is grounded in a series of key technologies, which can be conceptualized as a structured and multi-layered architecture. This architecture consists of a Perception Layer, which senses users and environments; a Decision-Making Layer, which processes information and determines appropriate actions; and a Generation and Execution Layer, which creates responsive and dynamic content. These layers operate in coordination, supported by robust learning models, to deliver intelligent and immersive experiences.

As illustrated in Figure 1, the framework adopts a flow-oriented structure that clearly delineates the process beginning with multimodal user input (e.g., visual, auditory, and physiological signals). The Perception Layer leverages CV and NLP to interpret data. The Decision-Making and Generation Layers utilize ML, **reinforcement learning (RL)**, and DL to enable intelligent reasoning and content creation. The system ultimately delivers intelligent VR experiences—such as smart virtual agents, dynamic virtual scenes, and personalized feedback—across key application domains including education, healthcare, gaming, and corporate training.

A critical feature of this framework is the feedback loop integrated into the system architecture. This loop channels user experience data back into the system’s input, effectively capturing the AI–VR system’s capacity for dynamic, continuous learning and self-optimization. The result is a complete and logically coherent closed-loop system that embodies the adaptive intelligence at the core of AI–VR integration.

2.1 Perception Layer: Enabling Accurate Insight in VR Environments

The Perception Layer endows VR systems with the ability to understand users and their environments through CV and NLP. Its unique value in VR lies in the following aspects:

- (1) *High-Precision Interaction and Rendering Optimization*: CV in VR surpasses basic image recognition. It provides sub-millimeter hand tracking, essential for high-risk applications like surgical simulation [10]. Additionally, gaze tracking supports foveated rendering, a technique that displays only the area where the user is looking in full resolution. This greatly decreases GPU load and addresses the main limitation of limited computing power in mobile VR devices.
- (2) *Immersive Dialogue Driven by Natural Language*: NLP in VR goes beyond basic voice command interpretation. It enables the development of Intelligent Virtual Agents capable of understanding conversational context and even recognizing user emotions [11]. These AI-powered virtual assistants [12] utilize technologies such as real-time voice-based sentiment analysis [13] to adapt their responses dynamically, significantly enhancing personalization and realism in interactions. In applications such as psychotherapy [14, 15] and customer service training [16], users can engage in meaningful, unscripted conversations with virtual characters. Furthermore, designing human-like interaction strategies to increase social presence [17] and user trust [18] has become a central focus of research in this field.

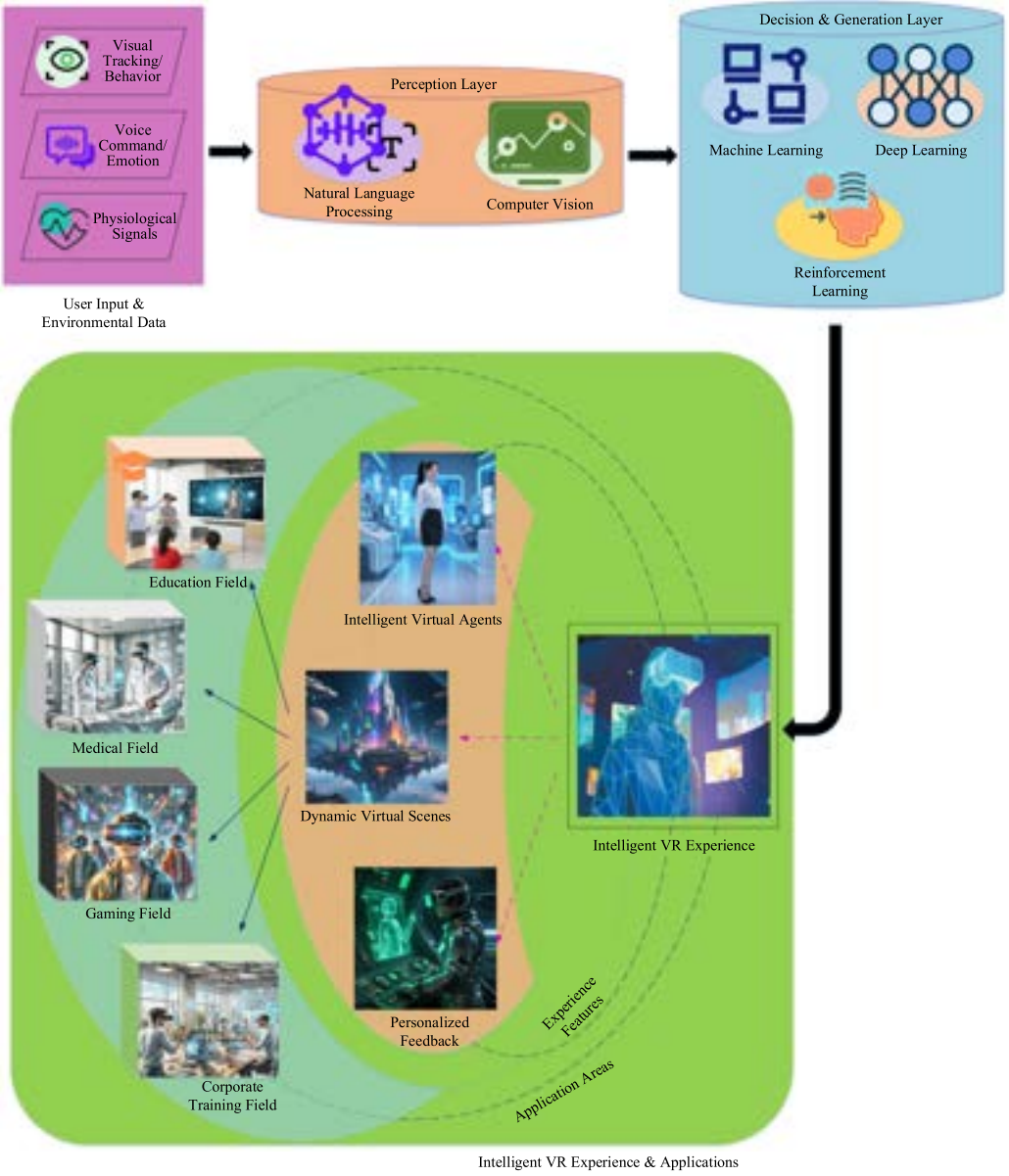


Fig. 1. AI-VR integration framework centered on Perception, Decision-Making, and Generation.

2.2 Decision-Making and Generation Layer: Constructing Dynamic and Intelligent Virtual Worlds

This layer leverages ML, RL, and DL to transform perception data into adaptive and dynamic VR experiences. Its value is reflected in the following aspects:

- (1) *Personalized Adaptive Learning*: In educational VR applications, ML models analyze student behavioral data—such as response times and gaze fixation points—to dynamically adjust the

difficulty and pacing of instructional content, thereby enabling truly adaptive learning. This addresses the inefficiency of traditional “one-size-fits-all” teaching methods [19–21]. When combined with principles from cognitive neuropsychology, the system can also perform adaptive assessments to further improve the learning experience [22]. Additionally, the integration of AI-based emotion recognition modules enhances the system’s contextual awareness, allowing it to better respond to users’ emotional needs [23].

- (2) *Real-Time Feedback and Dynamic Scene Generation*: RL’s dynamic reward mechanism is crucial in VR-based training. For instance, in pilot training, an RL agent can produce sudden weather changes based on real-time user actions and provide immediate feedback and rewards depending on response accuracy. This effectively addresses issues of delayed feedback and static scenarios found in traditional simulation training. In multi-agent systems, combining RL with federated learning strategies is vital for improving delivery efficiency in mobile VR networks [24]. In the gaming industry, **deep reinforcement learning (DRL)** is commonly used to enhance AI models that respond dynamically to player behavior and emotional patterns [25]. Additionally, DL techniques—especially Generative Adversarial Networks and Neural Radiance Fields—are employed for procedural content generation, enabling the real-time creation of new, coherent 3D scenes and tasks based on user activity [26]. This expands possibilities for open-world games and exploratory training while also supporting **augmented reality (AR)** applications in areas such as smart cities [27, 28] and medical diagnostics [29], through the dynamic generation of high-quality virtual environments.

This approach tightly links the functional capabilities of AI technologies with their specific applications and problem-solving roles in VR, clarifying their operational logic across various domains, including interaction, learning, training, and content generation. Furthermore, the framework supports broader applications in user behavior prediction and affective computing. For instance, in VR-based psychotherapy, AI platforms can analyze user emotions to deliver personalized interventions [30], while in serious games and rehabilitation, emotion-sensitive systems enhance the effectiveness of health interventions [31].

A comprehensive synthesis of the reviewed studies is presented in Table 1, while the interconnections among these works are visualized in Figure 2.

The integration of AI across multiple application domains advances the intelligence and adaptability of VR environments, enabling the development of personalized and interactive experiences. This study provides a comprehensive examination of AI-driven VR applications in education, gaming, healthcare, and corporate training, illustrating the multifaceted contributions of AI to the evolution of virtual environments.

In summary, by developing a “perception–decision–generation” technical framework, this section systematically explains the foundational AI technologies in VR and their key functional roles. After establishing this comprehensive technical and application blueprint, the next section builds on this foundation with a detailed thematic analysis. This transition reflects a step-by-step approach—from designing the architecture to exploring specific domains and from technical framework development to empirical validation. Specifically, Section 3 examines important sectors such as education, healthcare, gaming, and corporate training. By analyzing the latest research cases in these fields, it shows how the discussed framework is practically implemented, integrated, and validated in real-world situations, providing a clear and comprehensive view of AI–VR integration in practice.

3 State of the Art: Cross-Domain Applications Review

The convergence of AI and VR has progressed across various fields, including education, gaming, healthcare, and corporate training. Research and practical applications demonstrate that AI-powered

Table 1. Summary of Relevant Studies

Application Domain	Representative Studies	Perception Layer Technologies (CV/NLP)	Decision-Making and Generation Layer Technologies (ML/RL/DL)	Core Contributions
Intelligent Interaction and Virtual Agents	Uddin	General NLP interaction framework	–	Enhanced real-time NLP applications
	He et al.	Voice command interpretation	RL-based behavioral guidance	Improved AI assistant efficiency and intelligence
	Zadeh and Alaeifard	Speech emotion recognition	DL-based adaptive interaction	Enhanced interaction intelligence and personalization
	Grodniwicz and Hohol, Lee et al.	Voice interaction analysis	AI dialogue models	Explored AI applications in psychotherapy
	Malodia et al.	Voice interaction analysis	Optimized AI dialogue flows	Improved customer service experience and satisfaction
	Munnukka et al.	Interaction behavior and attitude analysis	–	Uncovered the value of anthropomorphic design
	Xiong et al.	–	ML-based trust model evaluation	Optimized personalization and user trust
Personalized Education and Training	Gligorea et al., Sajja et al., Demartini et al.	Learning behavior data analysis	ML-based personalized learning paths	Provided theoretical and practical frameworks for adaptive learning
	Halkiopoulous and Gkintoni	BCI-based cognitive state detection	Cognition-adaptive assessment	Explored cognition-driven VR learning models
	Shi	Multimodal emotion detection	Emotionally adaptive learning	Highlighted the importance of context awareness
	Liu et al.	Network condition sensing	RL-based network optimization	Enhanced efficiency of VR content delivery
	Hu et al.	Player behavior and emotion analysis	DRL-based adaptive game decision-making	Enabled adaptive AI game interaction
Dynamic Scene and Content Generation	Park et al.	–	DL-based procedural content generation (PCG)	Strengthened security and immersion in the metaverse
	Almatared et al., Heidari et al.	CV-based digital twins visualization	DL-based dynamic scene generation	Enabled adaptive VR environments for smart cities
	Bai et al.	CV-based 3D visualization of medical data	DL-based dynamic medical scene generation	Improved training and diagnostics in healthcare
User Behavior Prediction and Affective Analysis	Sadeh-Sharvit et al.	Multimodal emotion and behavior analysis	AI-based personalized intervention	Verified AI effectiveness in psychological intervention
	Tolks et al.	Physiological signal-based emotion sensing	RL-based optimization of health interventions	Enhanced health outcomes in serious games using VR

VR systems not only enhance interaction and immersion but also deliver intelligent, adaptive, and context-aware solutions tailored to various scenarios. This section explores the current state of AI–VR integration, highlighting its role in education and training, gaming and entertainment, healthcare and psychotherapy, as well as corporate training and simulation.

3.1 AI-Empowered VR in Education and Training

The integration of AI and VR in the field of education has garnered increasing attention from researchers, resulting in comprehensive investigations that span theoretical frameworks, technical implementations, and empirical validations. As a result, a clear research trajectory has emerged (Figure 3).

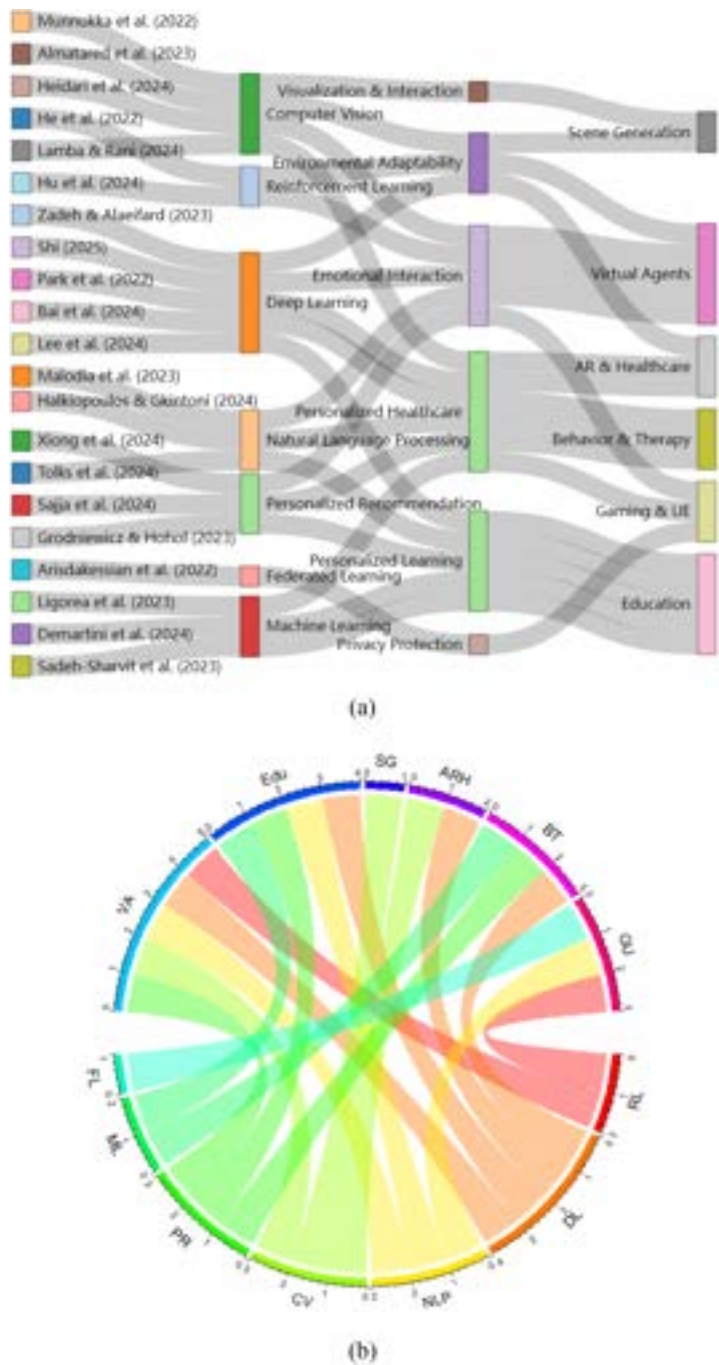


Fig. 2. Relationship analysis of the reviewed literature ((a) Sankey diagram; (b) Chord diagram).



Fig. 3. Relationship strength diagram among the reviewed studies.

Early studies provided a strong theoretical foundation for this interdisciplinary field. For example, Maghsudi et al. [32] explored the enormous potential of personalized education in the AI era, highlighting the need for data-driven strategies to improve learning experiences. Later, Hashim et al. [33] performed a systematic review of the literature to identify key technological trends and application models of AI and VR in personalized learning, mapping out a broad research landscape. Zohuri and Mossavar-Rahmani [34] further investigated the collaborative relationship between AI and VR, theorizing how their integration could transform the future of education. Together, these studies laid the theoretical groundwork for AI-VR applications in education.

Building upon this theoretical groundwork, subsequent research has focused on translating conceptual frameworks into practical technological solutions. A representative example is the work of Tapalova and Zhiyenbayeva [35], who investigated how AI could plan personalized learning paths in VR to enhance students' autonomous learning abilities.

On a more technical level, Pratama et al. [36] explored how DL algorithms could optimize instructional content in VR environments, making it more interactive and adaptive. Meanwhile, Limu et al. [37] assessed the potential of large language models such as ChatGPT to serve as virtual tutors, validating their feasibility in delivering real-time feedback and personalized guidance. Guettala et al. [38] turned their attention to cutting-edge generative AI technologies, examining their unique advantages in producing immersive and adaptive learning content. These diverse technological approaches collectively advance the practical implementation of AI-VR educational systems.

The ultimate goal of both theoretical and technical developments is to empirically validate their effectiveness in real educational settings. In this regard, the study by Song et al. [39] provided a landmark example. The researchers developed an immersive learning platform named "Learning-verseVR," which integrated generative AI with VR to analyze students' emotional states in real time and recommend personalized curricula. Their comparative experimental results offered robust empirical evidence, demonstrating that the platform significantly enhanced students' emotional engagement and learning efficiency, thereby validating the practical value of AI-driven VR education.

In addition, large-scale adoption of such technologies depends on user acceptance. Lampropoulos et al. [40] specifically addressed this issue through a social survey that examined public attitudes toward AI/VR educational technologies. Their findings indicated that factors such as digital literacy and educational background played a critical role in influencing user acceptance.

Beyond general education, the integration of AI and VR also demonstrates considerable value in professional skills and vocational training (Figure 4). Li et al. [41] examined how AI and VR created intelligent virtual business environments in entrepreneurship education, significantly improving students' practical skills and innovative thinking. In the broader context of professional development, Bukartaite and Hooper [42] investigated the profound impact of AI and automation technologies on future skill requirements, emphasizing the critical role of AI–VR in adaptive learning and skill reinforcement.

Focusing on industrial applications, Holuša et al. [43] developed an AI-driven VR corporate training model, demonstrating that immersive systems can significantly reduce the time required to master complex technical skills. Aranda et al. [44] focused on employee training under Industry 4.0, illustrating how AI and VR simulated intricate industrial processes to help workers acquire new skills more efficiently and safely. Targeting technical labor, Mandal [45] and Amdan et al. [46] provided theoretical and empirical evidence, respectively, on the effectiveness of AI–VR training models in enhancing learning speed, adaptability, and instructional support for educators.

In summary, research in the education domain exhibits a clear progression from the construction of theoretical frameworks to diversified technological implementations, followed by rigorous empirical validation and analysis of social acceptance. These studies complement each other across multiple dimensions—not only covering general education but also extending into skill development and vocational training. Together, they constitute a comprehensive body of evidence that systematically reveals the substantial potential of AI–VR integration in advancing personalized, intelligent, and efficient education.

3.2 AI-Driven Gaming and Entertainment Experiences

The integration of AI and VR technologies is fundamentally reshaping the core interaction paradigms of the entertainment industry, enabling users to engage with multidimensional, dynamic experience systems. Unlike traditional VR content reliant on fixed scripts, algorithmically enhanced virtual spaces can iteratively adjust environmental parameters based on user interaction patterns. Bai et al. [47] developed a volleyball training simulator that integrated facial micro-expression recognition with motion capture technology. Their virtual training partner demonstrated continuous emotional state transitions, effectively reducing the psychological distance between skill acquisition and emotional engagement. Benlamine et al. [48] designed the BARGAIN system, which utilized behavioral and emotional rules to adjust VR gaming environments. The system presented predefined ethical dilemmas to trigger user decision-making while employing real-time feedback mechanisms to enhance virtual characters' social attribute expression. Zhang et al. [49] incorporated AI with fifth-generation (5G)/IoT technologies into VR/AR platforms, establishing a player behavior prediction model that provided personalized interaction optimization solutions for digital twins scenarios. Dobre et al. [50] developed a narrative engine that combines eye-tracking with electroencephalogram signal analysis. Their immersion evaluation algorithm dynamically adjusted story pacing, improving emotional interaction fluidity and player immersion.

These technological breakthroughs not only transformed content production paradigms but also influenced human cognitive evolution at the neuroplasticity level. Smerdov et al. [51] developed an AI prediction model based on multi-sensor data fusion. By analyzing player physiological indicators and operation trajectories, they built a performance prediction system that enhanced behavioral optimization in personalized recommendation systems. Lee and Kwon [52] constructed an adaptive

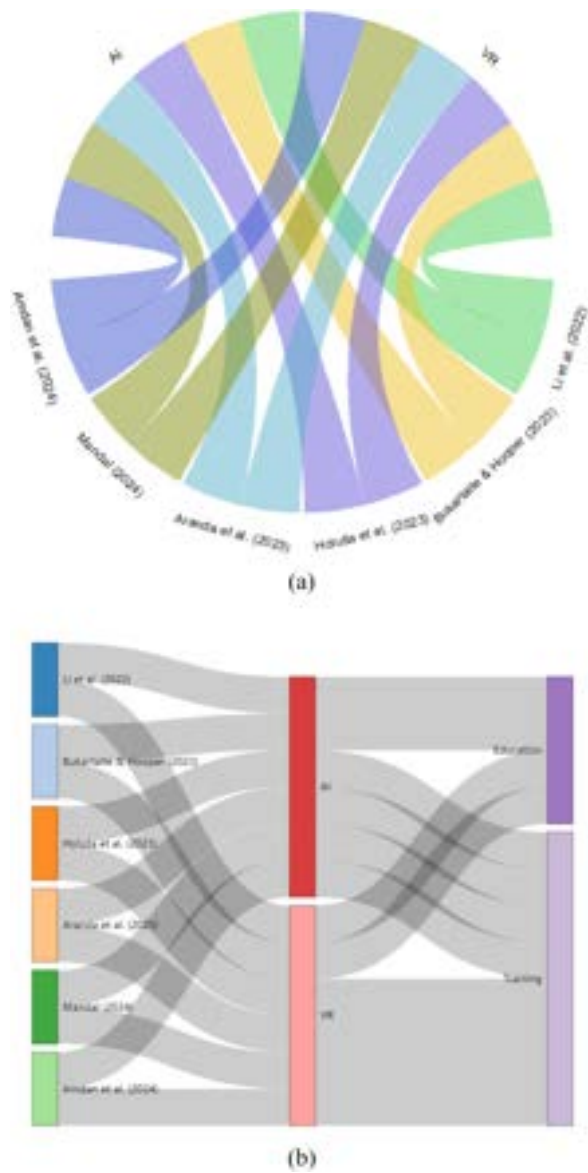


Fig. 4. Relationship diagram of applied technologies and research domains across studies ((a) Chord diagram; (b) Sankey diagram).

consistency model for shared object management in mixed reality environments. Their intelligent synchronization mechanism for distributed interaction scenarios effectively maintained system integrity during multi-user concurrent operations. Liu et al. [53] implemented a real-time response system for virtual characters in VR safety education scenarios. The system combined behavior prediction algorithms with emotion recognition modules, enabling multidimensional parameter iteration based on user feedback. Ojha et al. [54] designed a dynamic VR horror game framework that employed convolutional neural networks to analyze player posture features and heart rate

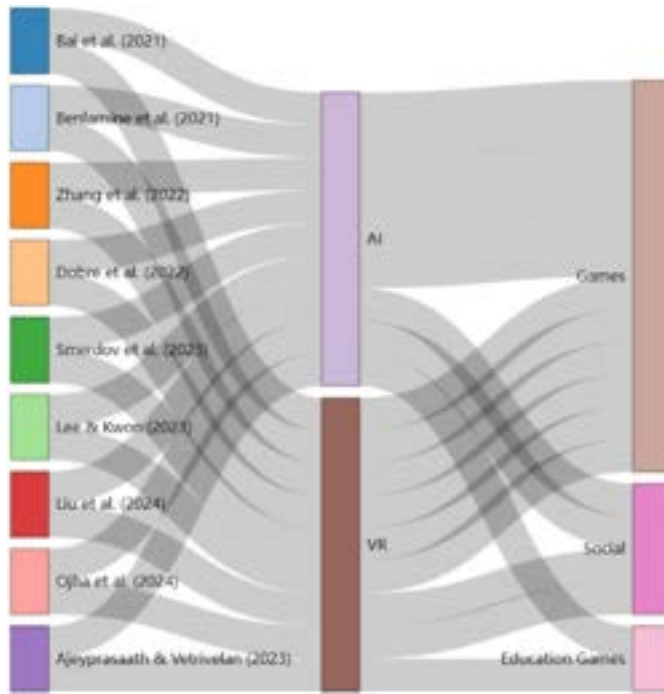


Fig. 5. Relationship diagram of technologies and domains used in the literature.

variability parameters. This allowed instant adjustment of narrative logic and challenge intensity. Ajeypasaath and Vetrivelan [55] created an ML-based Quality of Experience prediction framework. By decoding the mapping relationship between 5G network transmission characteristics and user gaze focus, they achieved intelligent configuration of video streaming parameters. The technological and domain analysis referenced in the studies is summarized in Figure 5.

Beyond intelligent virtual characters, AI plays a pivotal role in game narrative progression. The application of ML and affective computing allows AI systems to interpret player emotional states, dynamically adapting storylines and interaction modes to create more personalized gaming experiences. Cho et al. [56] proposed the DL-based asymmetric virtual environment framework as an exemplar of DL-VR integration. This framework aimed to establish asymmetric virtual environments to enhance metaverse immersion; It primarily employed adaptive content generation strategies, utilizing intelligent emotion perception and behavior prediction to achieve more natural virtual character interactions, thereby improving users' overall experience in immersive virtual spaces. Hu et al. [57] systematically reviewed DL applications in gaming, with particular focus on data-driven emotion analysis methods. Their research thoroughly investigated neural networks' potential for enhancing virtual characters' emotional expression and analyzed DL's role in non-player character's behavior modeling, player emotion feedback analysis, and adaptive gaming experience optimization. These insights have been offered with a lot of useful references for further game AI direction exploration.

Rokhsaritalemi et al. [58] integrated AI and geographic information system technology, using **extended reality (XR)** to explore emotion analysis mechanisms in urban services. They constructed an urban interaction framework that added more perception dimensions through VR and AR technologies. Iterative updates to the public service interface based on emotional feedback data

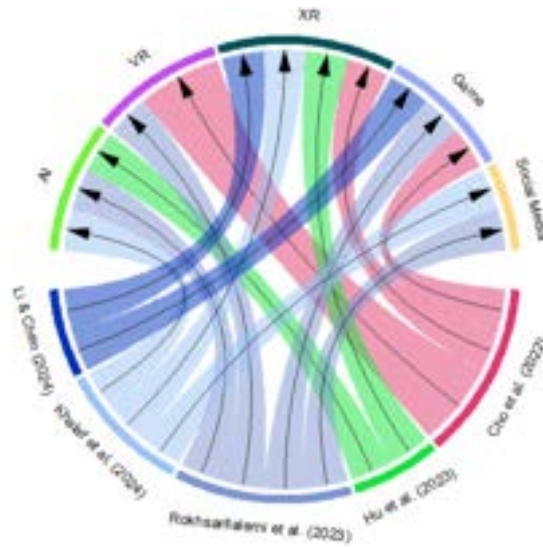


Fig. 6. Relationship diagram of literature [56–60] with technologies and application areas.

enabled efficient improvements in service efficiency and human-machine cooperation in smart cities. Khalaf et al. [59] developed a neuro-fuzzy emotion recognition model combining AI and VR technologies. The team used adaptive content-generation algorithms for real-time evaluation of users' emotional features; they adjusted AI behavioral logic to create environment elements that matched users' psychological expectations, achieving satisfactory experimental results that enhanced user immersion and emotional communication between virtual assistant characters and users. Li and Chen [60] applied text mining and sentiment analysis techniques to study the interaction mechanisms of virtual humans on social networks. They established NLP models to track users' emotional fluctuation patterns, discovering that the linguistic styles and emotional expression methods of virtual characters influence user experience. Consequently, they proposed various interactive optimization strategies for virtual characters, offering paradigm-shifting innovations in affective computing and behavior modeling. The relationships among the literature [56–60], related technologies, and application areas are depicted in Figure 6.

3.3 AI-Assisted VR in Medicine and Psychotherapy

The healthcare sector is actively exploring AI–VR convergence pathways. While VR constructs highly realistic environments for patient therapy and rehabilitation training, AI imbues these scenarios with dynamic decision-making and adaptive regulation capabilities. An integrated VR–AI system for physical rehabilitation was built by Bai et al. [61], specifically targeting stroke patients with continuous health management solutions. The system employed intelligent algorithms to dynamically adjust training parameters, enhancing recovery efficiency while strengthening remote monitoring capabilities.

Swarnakar and Yadav [62] analyzed AI's application value in motor function recovery, covering rehabilitation progress assessment, personalized training module generation, and recovery outcome prediction model construction. A study by Mennella et al. [63] revealed AI's multidimensional potential in future rehabilitation service systems through systematic literature analysis. They identified applications such as automated monitoring mechanisms, intelligent assisted training systems, and big data-driven personalized rehabilitation strategy design. Baranyi et al. [64] investigated

AI's supervisory and assistive roles in rehabilitation scenarios. They validated the synergistic effects of multimodal technologies. Their findings demonstrated that intelligent monitoring systems could simultaneously enhance training effectiveness and reduce therapists' workload. AI-driven virtual rehabilitation systems in community settings were systematically discussed by some researchers, namely Abedi et al. [65]; they highlighted the coupling relationships between remote monitoring systems, customized training programs, and AI intervention strategies. Wang et al. [66] conducted clinical validation of VR rehabilitation for spinal cord injury patients, demonstrating VR's effectiveness in improving motor function and increasing patient engagement levels.

This synergistic application unlocks novel possibilities for physical function rehabilitation. During clinical interventions, AI algorithms analyze patients' kinematic parameters and physiological indicators to dynamically adjust virtual environment interactions, maintaining optimal therapeutic intensity within safe thresholds. Nair and Sakthivel [67] developed a DL-based upper limb rehabilitation training system that analyzed movement data through algorithms, facilitating personalized adaptation and motion recognition accuracy. Maskeliūnas et al. [68] constructed the BiomacVR system by integrating depth sensors with VR technology, achieving precise posture and movement analysis during rehabilitation training to improve intervention accuracy. Ekambaram and Ponnusamy [69] reviewed AI-assisted physical therapy applications in post-injury rehabilitation, emphasizing ML's role in rehabilitation assessment, training program optimization, and personalized rehabilitation planning. Wei and Wu [70] systematically looked at wearable sensors combined with ML algorithms in rehabilitation training, exploring their potential for movement monitoring, real-time feedback, and training program optimization. Zhang et al. [71] investigated ML-integrated rehabilitation robots for lower limb training, introducing data-driven training strategies that improved both efficiency and safety in motor recovery. Ismail et al. [72] developed a hand movement prediction model utilizing neural networks to analyze hand motion data, thereby enhancing personalization and real-time feedback capabilities in rehabilitation training. An et al. [73] evaluated DL-driven rehabilitation applications in adolescents with hemiplegic cerebral palsy, demonstrating improvements in cardiopulmonary function, gait, balance, and daily activity performance. Zhang et al. [74] implemented convolutional gated recurrent neural networks to optimize VR-assisted ankle rehabilitation decision-making, advancing both the intelligence and personalization levels of rehabilitation training. The technological and application guidance referenced in these studies is depicted in Figure 7.

3.4 AI-Driven Corporate Training and High-Risk Simulation

The integration of AI and VR technologies in corporate training and high-risk environment simulations derives its true value from their high degree of industry-specific customization. Unlike "one-size-fits-all" solutions, AI-VR can address the unique pain points of specific sectors through precise simulation and targeted training. This section offers a sectoral analysis of this customization feature.

In asset-intensive industries such as smart manufacturing and high-tech agriculture, the primary goal of AI-VR is to optimize production processes and enable predictive management. Here, simulations focus not on emergency response training for personnel but on precise replication of equipment and environmental systems for data-driven analysis. For instance, Nagy et al. [75] examined the application of AI and virtual simulation algorithms in smart factories of small and medium-sized enterprises. Their solution centered on using DL to simulate time-series data of robotic systems' equipment operations, aiming to enhance production line efficiency rather than simply recreating the factory's physical appearance. Similarly, Rahman et al. [76] explored comparable applications in high-tech agriculture. They developed an intelligent greenhouse management framework driven by digital twins technology and ML, which enabled real-time monitoring and

In practical training scenarios, this emphasis on cognition becomes more evident. Johnson et al. [86] studied how explainable AI could enhance emergency medical response, particularly in opioid overdose treatment, by offering clear and trustworthy decision support for first responders. To integrate such capabilities into frontline operations, Damaševičius et al. [87] proposed the “Emergency Services IoT” framework, which used sensors and AI to enhance real-time monitoring and decision-making in high-risk contexts. At a broader level, Velez and Zlateva [88] discussed the data reliability and ethical challenges of using AI in disaster risk management. On the urban safety scale, Behdadfar and Samaei [89] investigated the development of city resilience assessment models using ML to support municipal-level emergency preparedness and response.

The industry-specific analysis above demonstrates that the applications of AI–VR technology are far from homogeneous. From time-series simulation for production optimization to enhancing cognitive decision-making in emergency responders, its true practical value lies in offering highly customized solutions tailored to the unique demands of different sectors. The relationship analysis of the literature in corporate training is visualized in Figure 8.

4 Challenges, Prospects, and Future Directions of AI–VR Integration

The ongoing evolution of AI within the VR domain has spurred significant innovations and developments. However, despite the substantial potential AI holds for enhancing VR applications, a range of challenges persists across technological, ethical, and user acceptance dimensions, as depicted in Figure 9. This study aims to further investigate the primary obstacles confronting AI in VR applications, offering insights into potential future advancements and directions.

4.1 Core Challenges of Technological Integration

The deep integration of AI and VR technologies presents a core challenge: achieving a balance among latency, accuracy, and cost—three inherently conflicting dimensions. Users’ pursuit of highly immersive experiences requires systems with high-precision, low-latency real-time responsiveness. However, this often conflicts with the high computational cost and data processing bottlenecks. Different technical approaches demonstrate varying performance orientations and application boundaries when addressing this fundamental contradiction.

4.1.1 Tradeoffs between Real-Time Data Processing and Edge Computing. One of the most immediate challenges in AI–VR applications is handling massive volumes of real-time data. For instance, the studies by Chen and Chen [90] and Aly et al. [91] focused on real-time facial expression recognition in VR environments for emotion analysis, which confronted significant computational latency and processing capacity limitations. To address this issue, edge computing has emerged as a solution, aiming to offload part of the computational tasks to user-proximal devices.

The edge computing framework for facial expression recognition proposed by Aikyn et al. [92] sought to improve the precision of real-time interaction. However, as highlighted by Wang et al. [93] in their study on XR streaming, edge nodes must carefully balance processing power and energy consumption, requiring algorithms to be highly lightweight. Moreover, real-time processing also raised security concerns. Sha et al. [94] investigated how AI techniques can be utilized to identify security vulnerabilities in VR devices.

4.1.2 Digital Twins Systems. Unlike edge computing, which emphasizes low latency, digital twins technology focuses on achieving high fidelity and accuracy in simulation. Lv et al. [95] explored the optimization of virtual environments using digital twins. However, this entailed detailed modeling of physical entities and continuous data synchronization, resulting in high computational and maintenance costs. Its key advantage lies in enabling complex offline simulations and predictions, which are critical for scenarios that prioritize result accuracy over real-time responsiveness.

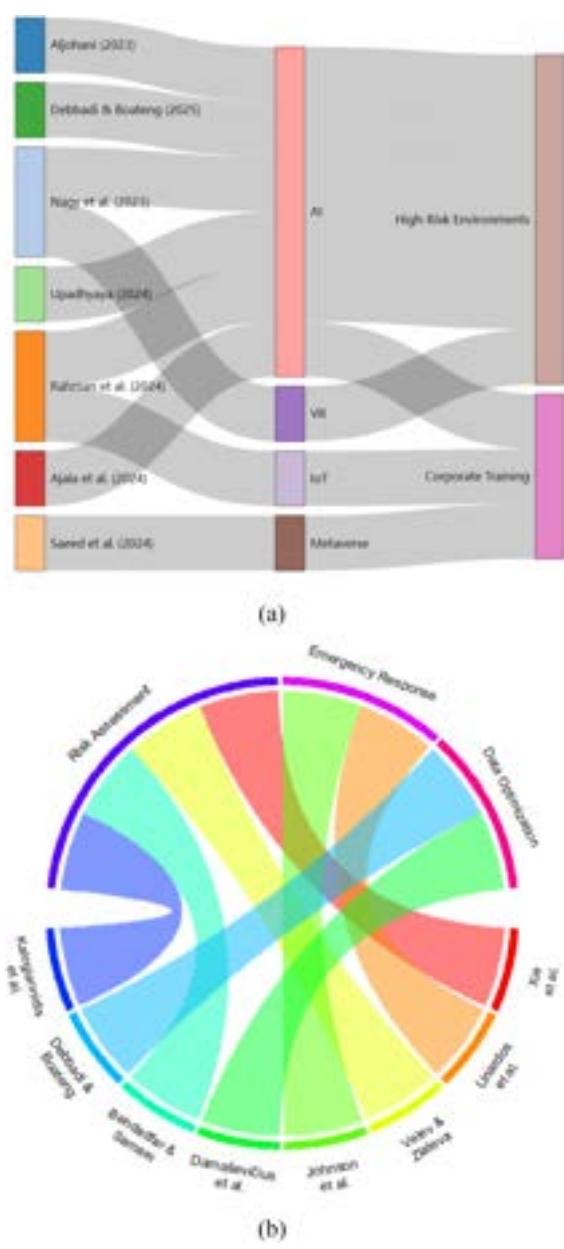


Fig. 8. Relationship analysis of literature in corporate training ((a) Sankey diagram; (b) Chord diagram).

4.1.3 Distributed Learning and Cross-Modal Integration. When AI-VR systems involve multi-user data processing or the integration of multiple AI capabilities, system complexity increases exponentially. Distributed or federated learning aims to harness decentralized resources for collaborative training. Although this approach protects user privacy, it also introduces substantial challenges related to model synchronization and communication overhead. A deeper challenge lies in cross-modal integration—how to seamlessly combine heterogeneous AI models such as CV,

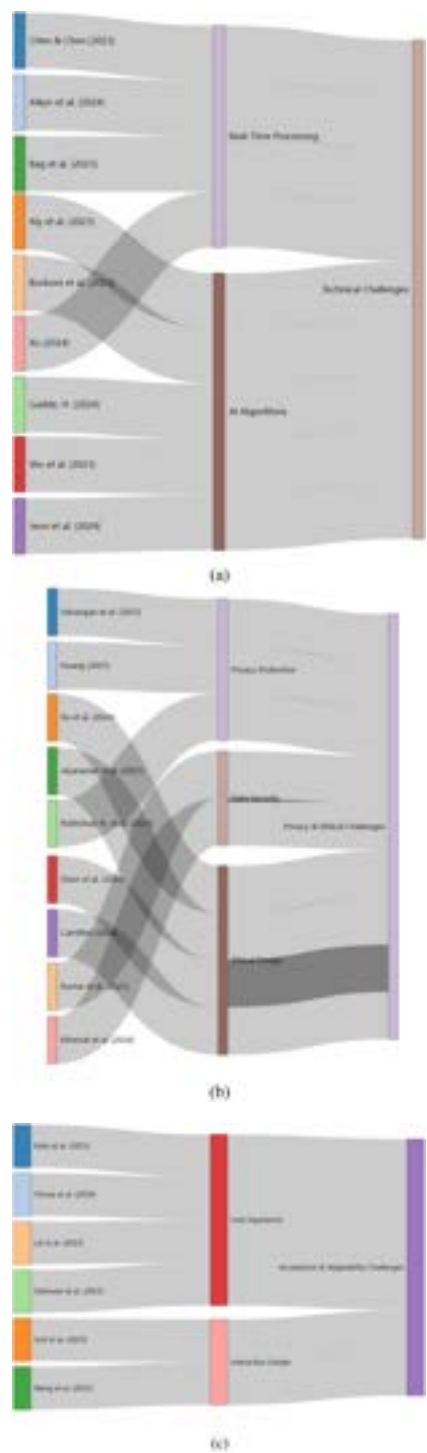


Fig. 9. Analysis of challenges in the literature ((a) technological challenges; (b) user privacy and ethical challenges; (c) user acceptance and adaptability challenges).

NLP, and RL. The study by Borboni et al. [96] revealed significant coordination failures between AI systems and electromechanical devices in industrial robotics, especially in the synchronization of temporal data and the nesting of decision logic. Similarly, Wu et al. [97] identified asynchronous interactions between operator behavior and AI decision logic in autonomous driving systems, which could cause delayed system responses. In emotion-driven collaborative robotics, Jeon et al. [98] pointed out the difficulty of quantifying emotional signals and mapping them to behavioral strategies. Additionally, at the database level, Gadde [99] revealed compatibility issues between ML models and traditional databases in terms of memory and processing cycles. Collectively, these studies underscore the significant engineering challenge of building a stable and efficient integrated AI-VR system.

4.1.4 Challenges of General AI and Generative AI. While general AI approaches offer broad application prospects [100,101], they also face challenges, including limited model adaptability and high training costs. In particular, generative AI, despite its great potential for personalized content creation [102], still requires significant improvement in terms of diversity, consistency, and real-time adaptability of generated content. In fields such as education [103] and sports training [104], ensuring that AI-generated feedback is both accurate and personalized remains an unresolved challenge.

In summary, the choice of technological pathway is not a matter of superiority, but rather a strategic tradeoff driven by specific application scenarios. For example, surgical simulations (more aligned with the digital twin approach) prioritize high precision and fidelity, accepting higher offline computational costs and tolerating some operational latency. In contrast, fast-paced multiplayer VR games (more reliant on edge computing) prioritize ultra-low latency, even at the expense of model complexity and environmental detail. Therefore, future research must explore hybrid architectures tailored to specific scenarios, aiming to achieve optimal integrated performance by understanding these fundamental tradeoffs. To systematically clarify these tradeoffs, this section presents a comparative analysis of key technological pathways in AI-VR applications, as shown in Table 2.

4.2 User Privacy and Ethical Dilemmas

The widespread adoption of AI-VR technologies has brought user privacy and ethical concerns to the forefront as critical challenges. AI systems in virtual environments routinely collect sensitive user data, such as behavioral patterns, emotional fluctuations, verbal interactions, and physiological feedback, creating significant leakage risks when this information undergoes non-standardized processing. Academic circles actively explore multidimensional protection frameworks through distributed encryption technologies and dynamic permission management mechanisms. Selvarajan et al. [105] established a lightweight blockchain-based AI security framework that achieved real-time data encryption and anonymous transmission protocols in industrial IoT scenarios. The study particularly emphasized the necessity of establishing dual verification mechanisms during cross-system collaborative computing to ensure attack resistance and information integrity in distributed storage nodes. The educational technology sector faced similar ethical dilemmas. Huang [106] demonstrated that a balance must be struck between personalized learning algorithms and student privacy rights. The application of DL models in educational virtual environments integrates access auditing functions and data desensitization strategies to prevent third-party institutions from reverse-engineering user identity characteristics through metadata analysis. Qayyum et al. [107] addressed the issues of security and trust in AI-XR, exploring the impact of AI applications in the metaverse on privacy and security. While they proposed an important security framework, the implementation of this technology in safeguarding user data and preventing malicious attacks

Table 2. Comparative Analysis of Key AI–VR Technological Pathways: Performance and Application Scenarios

Technological Pathway	Representative Studies	Performance Advantages	Core Challenges	Example Application Scenarios
Edge Computing	Aikyn et al., Wang et al.	Low latency, local data processing, reduced cloud burden	Limited computing and storage capacity at endpoints; difficulty in light-weight model design	VR gaming, real-time remote collaboration, instant gesture/facial expression interaction
Digital Twins	Lv et al.	High fidelity and precision; supports complex simulation and prediction	Demanding real-time data synchronization, high modeling, and computational costs	Industrial process simulation, precision surgical planning, urban planning
Distributed/Federated Learning	Liu et al.	Strong data privacy protection, high scalability, utilization of distributed resources	Complex model synchronization; high communication overhead among nodes	Multi-user personalized recommendations, large-scale user behavior analysis, collaborative AI training
Cross-Modal Fusion Algorithms	Borboni et al., Wu et al., Jeon et al.	Richer user experiences; more comprehensive understanding of user states	High system complexity; difficulty aligning multi-source data sequences	Affective computing, complex human–machine dialogue systems, intelligent cockpits
Real-Time Data Processing and Security	Chen and Chen, Aly et al., Sha et al.	Improved responsiveness and security	Computational latency, data bottlenecks, security vulnerabilities	Affective analysis in online education, VR device vulnerability detection, real-time tactical training in sports
General AI and Generative AI	Ribeiro de Oliveira et al., Lv, Xu	High generalizability; powerful content generation capabilities	Model adaptability, content consistency, high training costs	Intelligent assistant systems, personalized content generation, metaverse human–machine interaction

remains challenging, particularly in ensuring security within high-complexity XR environments. Xu et al. [108] proposed privacy-by-design principles for ML data collection, simultaneously improving user experience while enhancing privacy security. The critical challenge remained in achieving an equilibrium between effective data collection in ML processes and minimizing privacy leakage risks. Particularly in AI–VR integrated applications, real-time data collection and processing make user privacy protection especially crucial. A work by Kumar et al. [109] investigated explainable AI and blockchain applications in the metaverse, focusing on security and privacy protection issues. Privacy protection challenge involved dealing well with a balance between anonymity and data traceability within the application scenarios; within VR environments, it ensured personal data security from being stolen under virtual interaction situations, which emerged as key ethical problems regarding this technological convergence. Henceforward, future AI agents or VR systems have to creatively advance privacy protection and data governance toward following ethical norms and creating lasting, trustworthy relationships between them and their users, respectively.

Ethics is also a crucial part of AI–VR integration because AI can influence user emotions and behaviors, so designing and implementing virtual characters should adhere to certain ethical principles. Jeyaraman et al. [110] explored the potential applications of ChatGPT in medicine, education, and science research, along with the relevant ethical dilemmas. These included protecting the privacy of personal medical data from being illegally exposed when combining these technologies; ensuring **AI-generated content (AIGC)** does not become unethical or violate people’s rights after creation, among others. Addressing these challenges becomes essential when deploying virtual character applications in sensitive areas. The prospects and ethical issues of AI in oral maxillofacial plastic surgery were researched by Rokhshad et al. [111], focusing on whether AI should manage patient privacy and data security. Should AI make decisions transparently and objectively in complex cases? How can we ensure AI-driven treatments remain fair, despite differing opinions on AI use? Moreover, how can we prevent the negative health impacts and adverse consequences

associated with AI use? Sison et al. [112] emphasized human-centered AI and examined the ethical gray areas surrounding ChatGPT. Concerns about the credibility of AIGC raise important questions, especially regarding the risk of misinformation. Additionally, it is challenging to verify information transparency between AI systems and other entities. The core dilemma is ensuring that AI systems adhere to ethical standards throughout the information generation and dissemination processes in VR applications. Particularly in education and psychotherapy, VR virtual characters might cause discomfort or addiction among users. Therefore, experts and scholars have proposed creating a general moral code for AI and VR technologies to enhance their service in real-world situations. Specifically, developers must carefully consider potential negative consequences before designing and must take every possible measure to prevent technology abuse and protect users from significant psychological harm.

This fusion of AI-VR technologies raises many questions, but core conflicts always revolve around issues like digital identity reconstruction and blurring the boundaries of HCI from virtual to physical spaces. The fact remains that the number of daily online social interactions is increasing in cyberspace because VR fosters different types of behavior models; meanwhile, it can also easily confuse a person's true self-identity. Khreisat et al. [113] examined algorithmic bias in certain educational decision-making scenarios. Additionally, they analyzed conflicting relations concerning two critical perspectives: balancing students' privacy with the need for unbiased, algorithm-based fairness in this case. They also proposed implementing personalized learning systems in schools to develop new human-machine collaborative decision-making mechanisms among stakeholders in shared settings. Camilleri [114] highlighted responsibility-allocation mechanisms within government bodies of AI organizations that showed structural discrimination against large datasets. At the same time, it was found that the opaque operations of algorithms often obscure decision transparency. Future research should focus on addressing the dynamic balance and challenges of algorithmically automated educational processes, with a particular emphasis on the need for manual intervention to set threshold limits before software-based decisions are applied to humans. The technological evolution process requires the simultaneous development of user psychological assessment models, interdisciplinary ethics review committees, and virtual-physical interaction warning systems to monitor potential societal disruptions through multidimensional indicators.

4.3 User Acceptance and Adaptability Challenges

The widespread adoption of AI-VR integration heavily depends on user acceptance and technological adaptability. Although there have been significant breakthroughs in AI and VR technology, it remains challenging to achieve customized interaction methods and deep immersion. Kelly et al. [115] systematically reviewed related factors influencing user acceptance of AI from technology, psychology, and culture dimensions. According to these conclusions, they analyzed the determinants of user acceptance of AI-VR combined applications: perceived ease of use, system reliability, and personal privacy. At the same time, how to break through this barrier to improve user adaptability and confidence was also a problem worthy of exploration. Yilmaz et al. [116] constructed a generative AI acceptability scale, verified its moderating role of transparent technical design based on user cognitive mode characteristics. They explored in detail the importance of solving technical pain points such as interface adjustment and processing (the three major pains of interaction latency, error rate of personalized customization, and strength of data encryption) for VR environments. Goli et al. [117] empirically studied user acceptance of AI chatbots, finding that users' acceptance was affected by various trust, privacy protection, and technical usability. For AI-VR application software, adjusting and optimizing user interfaces and interactive experiences, reducing user technicality, and improving user reliability are key challenges for increasing acceptance and adaptability. Potential solutions should focus on developing explainable interaction

frameworks and reinforcing user agency through neural feedback mechanisms, thereby achieving synergistic optimization between technical sophistication and user experience.

HamlAbadi et al. [118] reviewed the application of BCIs in the metaverse, highlighting the potential of integrating BCI with AI. However, despite offering the possibility of enhancing user experiences, the technology still faces challenges related to signal noise, real-time processing, and user adaptability, which significantly affect the sense of immersion in VR. Ferdousi et al. [119] explored user perceptions of VR scene design, emphasizing the impact of design on immersion in learning environments. While user preferences were revealed, the application of AI in scene design remains limited by insufficient interactivity and personalization, making it difficult to achieve a fully personalized immersive experience. El Saddik et al. [120] examined the creative potential of generative AI in the metaverse, specifically its role in generating multimodal content. Although AI increases content creation efficiency, challenges related to realism and user personalization in VR still exist, especially regarding the quality and consistency of generated content. Lv et al. [121] discussed the use of DL in HCI, emphasizing its importance in smart virtual environments. Despite progress in automation and adaptability within AI-driven HCIs, factors like the computational complexity of models and the need for real-time responses remain key challenges for improving immersion and user experience.

Substantial individual differences have a significant influence on the adaptive performance of AI-VR technologies. Variations in people's cognitive states and needs require that technological upgrades consider necessary adjustments. Lai et al. [122] found that a mutual shaping relationship exists between adolescent social adaptation skills and AIs' help systems in school education; however, due to the complex changes in social support networks, the impact of learning in virtual space should not be overlooked. When combining AI with VR applications, the challenge is balancing educational outcomes with adolescents' social adaptability in virtual environments. The key question becomes how to utilize AI to enhance social support and facilitate adaptive learning while ensuring positive youth engagement with new technologies. Wang et al. [123] developed a hierarchical memory framework based on synaptic plasticity principles, enabling optimized allocation of cognitive resources during knowledge transfer. Further analysis identified the current challenge as allowing users to quickly adapt to changing virtual environments, while also ensuring continuous technological evolution aligned with user needs and ongoing ability development. Dahmani et al. [124] assessed the effectiveness of improved learning methods in global radiation prediction, highlighting AI's role in improving real-time data prediction accuracy. Due to individual differences and preferences, the ability to continuously adapt to the environment and meet diverse user requirements in a flexible manner remains one of the biggest challenges faced by designers today. Researchers should further study user behavioral patterns and develop corresponding intelligent adaptation algorithms. Thus, technological development can stay aligned with the growth of human cognition and achieve synchronized progress.

Subramanyam et al. [125] evaluated the impact of user-adaptive real-time point cloud streaming in VR remote communication, revealing the potential of AI in real-time data processing. Although the study demonstrated performance improvements, AI-optimized point cloud data still faced challenges in high-latency network environments, particularly in terms of reduced resolution and insufficient transmission stability. Jansen et al. [126] introduced the open source VR2Gather system, emphasizing its role in adaptive, multi-party real-time communication for collaborative social VR environments. While the study expanded multi-party interactions, the ability of AI to dynamically adjust virtual environments and user states remains a bottleneck, particularly concerning performance in high-concurrency scenarios. Kim et al. [127] explored the application of multi-viewpoint layout design in VR concert experiences, proposing enhancements to immersion through diverse audience perspectives. However, AI-related issues, such as latency, image quality,

and scene adaptability during real-time multi-viewpoint switching, remain unresolved, thereby impacting the fluidity and realism of the VR experience.

4.4 Future Prospects and Research Directions

The deep integration of AI and VR signals a future full of opportunities, but the path forward is far from smooth. Its healthy development depends not only on continuous technological breakthroughs but also on the ability to construct targeted solutions to the core challenges outlined above, while simultaneously establishing robust early-warning and response frameworks.

To begin with, tackling the challenge of technological integration—especially the bottleneck of real-time computing—requires future research to focus on hybrid architectures that combine federated learning with edge computing. This architecture allows computationally intensive tasks to be processed at the network edge, significantly reducing latency. Simultaneously, federated learning enables collaborative model training and optimization without compromising user data privacy. The study by Shaheen et al. [128] provided valuable insights in this area, exploring AI-enabled mobile edge computing. By employing balanced federated learning strategies, their approach achieved both lower computational costs and improved data processing balance, offering a promising solution to the computational load versus latency dilemma in VR applications. Additionally, in high-risk scenarios such as industrial control and autonomous driving simulation, circuit breaker mechanisms should be integrated into the technical architecture. These mechanisms could automatically switch the system to a safe mode when AI decisions deviate or when latency exceeds thresholds, thereby preventing catastrophic outcomes.

Second, in response to the challenges of user privacy and ethics, future solutions must go beyond traditional encryption methods toward more proactive and verifiable privacy protection technologies. A particularly promising approach is the development of collaborative verification schemes that combine differential privacy with blockchain. Differential privacy protects individual data by adding statistical “noise,” while blockchain’s immutability and decentralization offer a transparent and auditable record of data access and model training processes. Tekchandani et al. [129] provided a significant example in this field by proposing a secure collaborative model based on blockchain and differential privacy, which enables the safe analysis of large-scale IoT data without compromising privacy. This technological combination has the potential to provide “self-verifying” privacy protection capabilities for AI–VR applications. However, technology alone is not a complete solution. In sensitive areas like medical VR and psychological treatment, strict ethical review processes must be established and enforced to ensure that technological applications always prioritize human well-being.

Third, with respect to user acceptance and adaptability, future development must adopt a human-centered approach that prioritizes users’ cognitive and emotional experiences based on multimedia emotional data such as audio-visual, text, and voice [130–131]. This implies a shift from “functional realization” to “experience optimization.” For instance, future AI models should be capable of dynamically assessing and adapting to users’ cognitive load. In particular, for educational VR scenarios, the integration of real-time cognitive load monitoring systems should become a mandatory design feature. When the system detects cognitive overload or signs of attention dispersion, it should be able to automatically adjust the complexity and pace of information presentation to avoid negatively impacting learning outcomes due to overuse or misuse of technology.

In summary, the future development of AI–VR should follow a path that is challenge-driven, solution-oriented, and risk-controllable. By aligning cutting-edge technologies (such as federated learning, edge computing, differential privacy, and blockchain) with specific challenges, and supplementing them with safeguard mechanisms such as circuit breakers, ethical reviews, and cognitive

monitoring systems, it becomes possible to ensure that this powerful technological convergence remains on a trajectory that is safe, trustworthy, and beneficial to society.

5 Conclusion

This study offers a systematic review and analysis of AI and VR integration. By developing a technical framework, examining cross-domain applications, and analyzing key challenges, it ultimately suggests future development paths. The main contributions can be summarized on three levels. At the theoretical level, the study innovatively develops an integrated AI–VR technical framework based on the “perception–decision–generation” paradigm. This model provides a unified and systematic perspective for future research in this interdisciplinary area. At the methodological level, the study employs an interdisciplinary and multidimensional approach that connects technical principles, industry applications, and ethical issues. This breaks down the siloed structure seen in earlier research and fosters a more comprehensive understanding. At the application level, the study identifies specific, practical technological solutions for challenges such as real-time computing limitations and privacy-ethical concerns. For instance, it recommends architectures like “federated learning + edge computing” and “differential privacy + blockchain” to increase the study’s practical relevance and usefulness.

In summary, the integration of AI and VR is shifting from a simple functional overlay to a deeply interconnected and systematic symbiosis. AI is no longer just a “plugin” for VR but is becoming an “intelligent core” that is transforming the logic of virtual environment creation, interaction models, and application boundaries. Looking ahead, a truly intelligent virtual world will develop into an organic system capable of autonomous perception, understanding, decision-making, and dynamic content generation. This conclusion summarizes the key findings of the study and emphasizes the foundational role of AI in advancing VR from passive presentation to proactive intelligence.

Despite its broad scope, the study has certain limitations—most notably, its focus on the macro-level framework without exploring detailed algorithmic aspects. Future research could fill this gap by exploring several key areas: optimizing algorithms and improving efficiency (such as lightweight models); advancing innovations in HCI (especially the deep integration of affective computing and brain-computer interfaces); developing software-hardware systems collaboratively (with an emphasis on enhancing the “federated learning + edge computing” architecture); and building ethical and governance frameworks together. These directions not only directly address the main challenges highlighted in this study but also provide a clear pathway for the safe and trustworthy development of AI–VR technologies.

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Received 7 June 2025; revised 10 August 2025; accepted 9 September 2025